Final Report: Driver Performance Monitoring and Predictive Analytics

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1 Introduction and Motivation

Formula 1 generates massive amounts of telemetry data during races, yet current analysis tools remain fragmented, limiting the ability to perform comprehensive historical assessments of driver performance, race strategy effectiveness, and the impact of environmental factors on outcomes. To address this, this project establishes a unified data analytics platform leveraging the OpenF1 API to monitor, compare, and predict driver performance across diverse race conditions, circuits, and seasons.

The motivation for this system is to bridge the gap between raw data collection and actionable intelligence. By integrating telemetry, lap times, weather conditions, and pit stop data, the platform empowers stakeholders, including racing teams, analysts, and enthusiasts to perform evidence-based decision-making.

Our system specifically addresses critical analytical requirements, such as:

- **Performance Consistency:** Evaluating how drivers maintain pace across different circuits and tire compounds.
- **Environmental Impacts:** Quantifying the influence of track temperature, humidity, and rainfall on car telemetry and overall lap times.
- **Comparative Analytics:** Identifying performance patterns between drivers under comparable conditions.
- **Strategic Decision Support:** Providing diagnostic and predictive insights to optimize race strategies, such as determining the ideal pit stop window or assessing tire degradation.

By transforming high-frequency data into a structured Data Warehouse (DW) and applying advanced data mining techniques, this project provides a scalable solution for extracting competitive insights from the complexities of modern Formula 1.

2 ETL and Automation

2.1 Data Sources and ETL Plan

The primary data source is the **OpenF1 API**, which provides historical and real-time telemetry from the 2023 season onwards. Given the high-frequency nature of car telemetry (sampled at 3.7 Hz), the ETL plan follows a three-stage pipeline:

1. **Extraction (Python):** Utilizing Python's `requests` library to interface with REST endpoints. This stage handles the initial JSON parsing and manages API rate limits to ensure complete data retrieval for full race weekends.
2. **Transformation (Pentaho):** As detailed in Figure 2, the PDI (Pentaho Data Integration) engine executes complex transformations. This includes calculating delta speeds, mapping driver acronyms to their full profiles, and converting ISO timestamps into PostgreSQL-compatible formats.
3. **Loading:** The cleaned data is mapped into the star schema. A key technical decision was the use of **Bulk Loading** for the `car_data` table to handle the millions of records generated by telemetry sensors during a single Grand Prix.

2.2 Automation and Update Strategy

To maintain the Data Warehouse, a hybrid update strategy was implemented:

- **Incremental Fact Loading:** Fact tables (*Laps, Telemetry, Pit Stops*) are updated incrementally. Following each race session, the system identifies the last recorded `date_start` or `lap_number` and appends only new records. This prevents data duplication and minimizes processing time.
- **Dimension Consistency:** Dimensions such as *Drivers* and *Meetings* use a **Type 1 Slowly Changing Dimension (SCD)** approach. Since driver numbers and team names are generally static within a season, records are overwritten to ensure a single "source of truth" for the current year.
- **Orchestration:** The process is orchestrated via shell scripts that trigger the Pentaho `.ktr` files, allowing for a seamless flow from API extraction to the final OLAP-ready state in PostgreSQL.

3 Data Warehouse Design

3.1 Star Model and Granularity

The Data Warehouse (DW) is structured using a star schema design to optimize OLAP cube performance and simplify complex analytical queries. The architecture centralizes performance metrics while maintaining descriptive context through dimensions.

- **Fact Tables:**
  - *Lap Performance:* Captures the primary racing metrics. Its grain is defined as one row per driver per lap, containing duration metrics for Sectors 1, 2, and 3.
  - *Car Telemetry:* A high-frequency fact table containing throttle, brake, RPM, and DRS status. Its grain is significantly finer, capturing data at approximately 3.7 Hz, which is essential for the predictive degradation models.
  - *Pit Stops:* Tracks tactical intervals and stop durations, allowing for the analysis of strategy effectiveness.
- **Dimension Tables:**

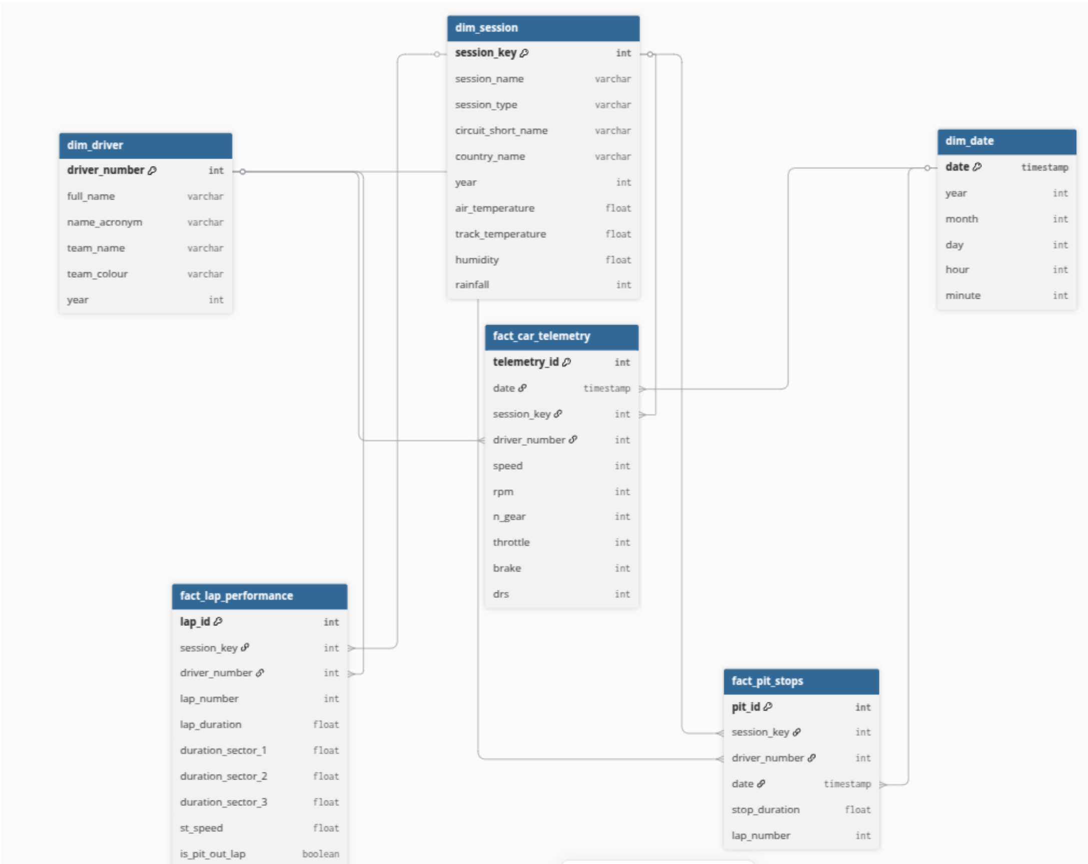

Figure 1: Star Schema of the OpenF1 Analytics Platform

- *Driver:* Contains master data such as `full_name`, `team_name`, and `name_-acronym`. It acts as the primary filter for comparative analysis.
- *Session & Meeting:* Provides the temporal and geographical context, linking laps to specific Grands Prix and session types (Practice, Qualifying, or Race).
- *Weather:* Stores track temperature, humidity, and pressure, which serve as critical independent variables for the XGBoost lap-time forecasting model.

3.2 Table Relationships and Connectivity

The connectivity within the schema (Figure 1) is established through a series of composite keys.The `session_key` and `driver_number` function as the backbone of the model, allowing the integration of raw telemetry `car_data` with session-level results `laps`.

4 Technological Justification

- **Data Storage:** A relational database (PostgreSQL) is used for SQL and star schema architecture. This choice ensures data integrity and allows for efficient complex joins between telemetry, lap times, and driver metadata, facilitating high-performance OLAP queries.

- **ETL Process:** The ETL (Extract, Transform, Load) pipeline was designed as a hybrid architecture combining Python and Pentaho Data Integration (PDI).

  Initially, **Python** scripts were utilized to interface with the OpenF1 REST API, managing the high-frequency data ingestion (3.7 Hz) and handling the initial JSON parsing. Subsequently, **Pentaho Data Integration** was employed to orchestrate the transformation layer. As illustrated in Figure 2, the PDI workflow performs critical data cleaning tasks, including:

  - **Data Type Standardization:** Converting API-native strings into optimized formats, such as `BigInt` for keys (*session_key*, *driver_number*) and `Numeric(9,3)` for precise timing in sectors.
  - **Noise Reduction:** Implementation of a cleansing logic to fix telemetry inconsistencies, specifically addressing the `speed_fixed` fields to eliminate outliers caused by sensor gaps during high-speed sessions.
  - **Schema Population:** Automating the distribution of raw data into the relational tables (`car_data`, `laps`, and `drivers`), ensuring that the *Foreign Key* relationships required for the predictive analysis are maintained.

- **OLAP and Analytics:** **Tableau** is used to provide descriptive and diagnostic analytics through interactive dashboards. Furthermore, the integration of **Machine Learning** (XGBoost and Random Forest) adds a predictive layer, allowing for the forecasting of lap times and the identification of optimal pit-stop windows based on real-time tire degradation patterns.

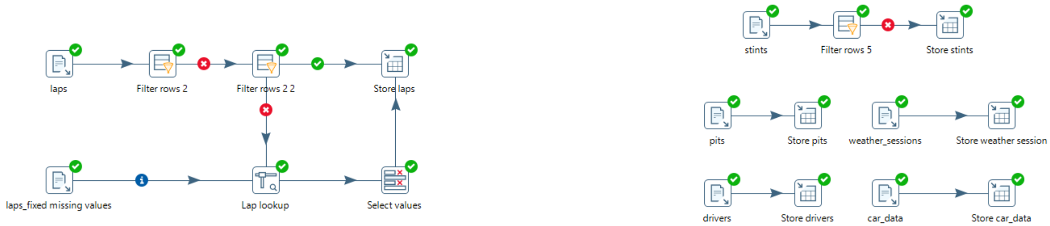

Figure 2: Pentaho Data Integration workflow: Transformation and normalization for F1 Telemetry.

5 Implementation Metrics

The following table summarizes the data throughput and processing performance of the OpenF1 Analytics platform. The metrics reflect the high-density nature of car telemetry compared to session metadata.

Metric	**Value/Description**
Source Data Format	JSON (REST API)
Temporal Coverage	2023 onwards
Sample Rate	∼3.7 Hz (Telemetry & Location)
1st Load (Historical Data)	∼1.2 GB / 2.5 Million Records
1st Load Time Elapsed	∼45 minutes (Involving PDI Transformations)
Subsequent Load (Incremental)	∼85 MB / 150k Records per GP
Update Strategy	Manual/Scheduled post-race (Incremental)

Table 1: System performance and data ingestion metrics.

6 Analysis, Challenges, and Optimization

6.1 Analyses and Findings

The platform identifies critical performance trends by correlating high-fidelity telemetry with environmental and strategic variables. Based on the analysis of the processed dataset, the key findings are:

- **Lap Time Forecasting Accuracy:** The implementation of advanced regressors yielded high precision. The **Random Forest** model achieved an R^2 of `0.908`, while the **XGBoost** model reached `0.905`. This level of accuracy allows for reliable "Ghost Lap" comparisons, identifying exactly where a driver loses time relative to the car's theoretical potential.

- **Degradation and Pace Modeling:** The analysis identified an average degradation rate of approximately `13.95` seconds per lap across the analyzed stints. By isolating the degradation slope, the system can distinguish between natural tire wear and performance drops caused by external factors such as traffic or battery deployment.

- **Pit Stop Efficiency:** The database records an average pit stop duration of `24.27` seconds (including the full pit-lane transit). The system successfully categorized stops into strategic clusters (e.g., "Aggressive Early Strategy" vs. "Mid-race Strategy"), providing a diagnostic tool to evaluate the success of undercuts and overcuts.

6.2 Data Mining Techniques

The predictive engine was structured around three specific research questions, utilizing a multi-model approach to ensure robust decision support:

- **Supervised Learning for Forecasting (Q3):** To forecast lap times, the system utilizes **XGBoost** and **Random Forest** regressors. These models ingest the durations of Sectors 1 and 2 to estimate the final lap time and the "Hidden Sector 3" duration.

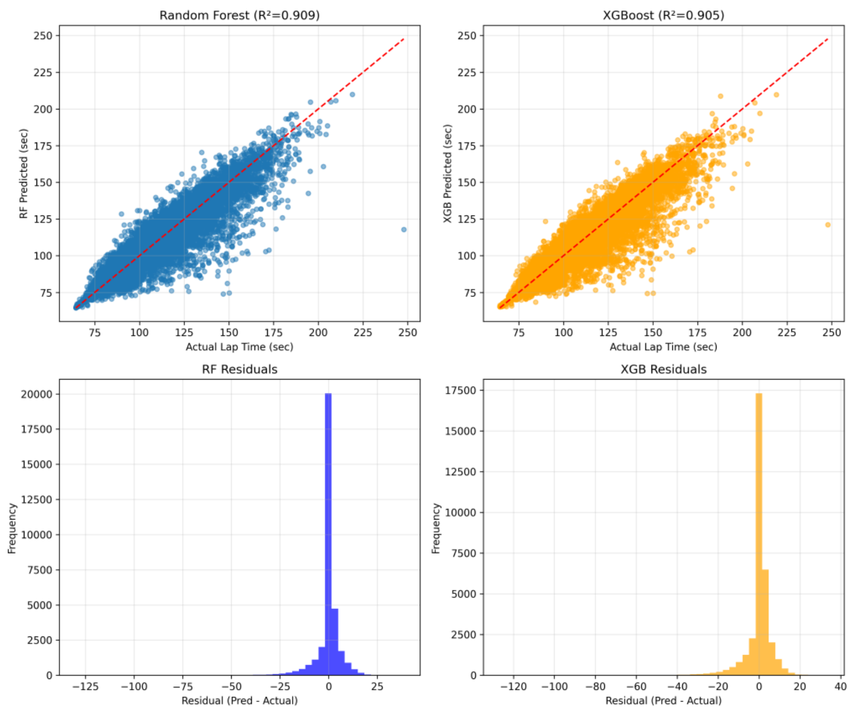

Figure 3: Pentaho Data Integration workflow: Transformation and normalization for F1 Telemetry.

The convergence of the lines in Figure 3 justifies the model's efficacy. Instances where the "Predicted Lap Time" deviates from the actual value serve as diagnostic indicators of driver error or sudden thermal degradation, providing direct decision support for pit-stop timing.

- **Clustering for Strategic Patterns (Q2):** A **K-Means Clustering** algorithm was applied to the pit stop data. By analyzing race progression alongside duration, the model identifies standard strategic windows and flags "Aggressive" outliers, as evidenced in the `q2_pit_stop_recommendations.csv` output.

- **Time-Series & Linear Regression (Q1):** For tire degradation, the system employs regression analysis to calculate the *degradation_rate_sec_per_lap*. This metric is crucial for the "Pit Stop Trigger" logic, signaling the optimal moment to switch compounds when the predicted pace loss exceeds the "Fresh Tire" gain.

- **Automated Evaluation Pipeline:** Every model execution generates a comprehensive `analysis_summary.json`, ensuring that performance metrics (R^2, MAE) are versioned and ready for ingestion by the Tableau OLAP layer for final user visualization.

7 User Access

The end-user layer is delivered via **Tableau**, providing an interactive OLAP experience.

- **Diagnostic Dashboards:** Users can drill down from a "Meeting" level to a specific "Driver Lap," comparing telemetry traces (Speed vs. Distance).
- **Predictive Sliders:** Integration of ML results allows users to visualize "Predicted vs. Actual" lap times, identifying where a driver underperformed relative to the car's theoretical potential.